Test of the overarrows package with non-standard subscripts (package altsubsup)

Julien Labbé

October 19, 2025

1 Subscript detection enabled

```
\TestOverArrow*{\withsubdetection}

$ \withsubdetection{\imath}_{sub} \qquad \withsubdetection{\imath}_[altsub]$
```

Test of `\withsubdetection` and `\withsubdetection*` macros

`\withsubdetection` for different math styles

`\displaystyle`	`\textstyle`	`\scriptstyle`	`\scriptscriptstyle`
$\overrightarrow{v}$	$\overrightarrow{v}$	$\overrightarrow{v}$	$\overrightarrow{v}$
$\overrightarrow{AB}$	$\overrightarrow{AB}$	$\overrightarrow{AB}$	$\overrightarrow{AB}$
$\overrightarrow{\text{grad}}$	$\overrightarrow{\text{grad}}$	$\overrightarrow{\text{grad}}$	$\overrightarrow{\text{grad}}$
$\overrightarrow{my\ long\ vector}$	$\overrightarrow{my\ long\ vector}$	$\overrightarrow{my\ long\ vector}$	$\overrightarrow{my\ long\ vector}$

`\withsubdetection` kerning

$$\overrightarrow{t}_{\overrightarrow{u}_{\overrightarrow{v}}} \qquad \overrightarrow{\imath}_0 \qquad \overrightarrow{v} = \overrightarrow{v}_x + \overrightarrow{v}_y + \overrightarrow{v}_z = v_x\,\overrightarrow{\imath} + v_y\,\overrightarrow{\jmath} + v_z\,\overrightarrow{k}$$

`\withsubdetection*` kerning

$$\overrightarrow{t}_{\overrightarrow{u}_{\overrightarrow{v}}} \qquad \overrightarrow{\imath}_0 \qquad \overrightarrow{v} = \overrightarrow{v}_x + \overrightarrow{v}_y + \overrightarrow{v}_z = v_x\,\overrightarrow{\imath} + v_y\,\overrightarrow{\jmath} + v_z\,\overrightarrow{k}$$

$$\overrightarrow{\imath}_{sub} \qquad \overrightarrow{\imath}_{\text{altsub}}$$

2 Subscript detection disabled

```
\TestOverArrow*{\nosubdetection}

$ \nosubdetection{\imath}_{sub} \qquad \nosubdetection{\imath}_[altsub]$
```

Test of \nosubdetection and \nosubdetection* macros

\nosubdetection for different math styles

\displaystyle	\textstyle	\scriptstyle	\scriptscriptstyle
$\overrightarrow{v}$	$\overrightarrow{v}$	$\overrightarrow{v}$	$\overrightarrow{v}$
$\overrightarrow{AB}$	$\overrightarrow{AB}$	$\overrightarrow{AB}$	$\overrightarrow{AB}$
$\overrightarrow{\mathrm{grad}}$	$\overrightarrow{\mathrm{grad}}$	$\overrightarrow{\mathrm{grad}}$	$\overrightarrow{\mathrm{grad}}$
$\overrightarrow{my\ long\ vector}$	$\overrightarrow{my\ long\ vector}$	$\overrightarrow{my\ long\ vector}$	$\overrightarrow{my\ long\ vector}$

\nosubdetection kerning

$\overrightarrow{t}_{\overrightarrow{u}_{\overrightarrow{v}}} \qquad \overrightarrow{\imath}_0 \qquad \overrightarrow{v} = \overrightarrow{v}_x + \overrightarrow{v}_y + \overrightarrow{v}_z = v_x\,\overrightarrow{\imath} + v_y\,\overrightarrow{\jmath} + v_z\,\overrightarrow{k}$

\nosubdetection* kerning

$\overrightarrow{t}_{\overrightarrow{u}_{\overrightarrow{v}}} \qquad \overrightarrow{\imath}_0 \qquad \overrightarrow{v} = \overrightarrow{v}_x + \overrightarrow{v}_y + \overrightarrow{v}_z = v_x\,\overrightarrow{\imath} + v_y\,\overrightarrow{\jmath} + v_z\,\overrightarrow{k}$

$\overrightarrow{\imath}_{sub} \qquad \overrightarrow{\imath}_{\mathrm{altsub}}$

3 Compatibility with **altsubsup** package

```
$ \vv{\imath}_{sub} \qquad \vv{\imath}_[altsub]$
```

$\overrightarrow{\imath}_{sub} \qquad \overrightarrow{\imath}_{\mathrm{altsub}}$